MAT 535 HW03

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Problem 1 (§17.1#10a,d)**.** (a) Prove that an arbitrary direct sum $\bigoplus_{i\in I} P_i$ of projective modules P_i is projective and that an arbitrary direct product $\prod_{j\in J} Q_j$ of injective modules Q_j is injective.

(d) Prove that $\operatorname{Tor}_n^R(A, \bigoplus_{j\in J} B_j) \cong \bigoplus_{j\in J} \operatorname{Tor}_n^R(A, B_j)$ for any collection of R-modules $\{B_j\}_{j\in J}$.

Solution. (a) Let $\{P_i\}_{i\in I}$ be a collection of projective modules, $u\colon M \to N$ is an epimorphism, and $f\colon \bigoplus_{i\in I} P_i \to N$. Define the canonical inclusion $q_i\colon P_i \to \bigoplus_{i\in I} P_i$ and let $f_i\colon P_i \to N$ be $f_i = f \circ q_i$. Since the P_i are projective, there exist $\tilde{f}_i\colon P_i \to M$ making the diagram

$$\begin{tikzcd}
& P_i \arrow[dl, dashed, "\exists \tilde{f}_i"'] \arrow[d, "f_i"] & \\
M \arrow[r, "u"] & N \arrow[r] & 0
\end{tikzcd}$$

commute. By the universal property of the coproduct, the maps $\tilde{f}_i$ factor through the q_i, i.e. there is a unique morphism $\tilde{f}\colon \bigoplus_{i\in I} P_i \to M$ such that $\tilde{f} \circ q_i = \tilde{f}_i$. Moreover, the universal property of the coproduct also says that f is uniquely characterized by the equations $f_i = f \circ q_i$. Since $(u \circ \tilde{f}) \circ q_i = f_i$ for all i, this implies that $u \circ \tilde{f} = f$, so $\tilde{f}$ is the desired lift of f.

The analogous result for injective modules follows from passing to the opposite category, or equivalently making the same argument but with the universal property of the product instead (since in that setting there would be maps $g_j\colon N \to Q_j$ which need to be extended to maps $\tilde{g}_j\colon M \to Q_j$ in a manner compatible with a given monomorphism $u\colon N \to M$).

(d) Fix a projective resolution $\cdots \to P_{j,2} \to P_{j,1} \to P_{j,0} \to B_j \to 0$ for each $j \in J$. Since direct sums of projective modules are projective by (a) and direct sums of exact sequences are exact, we obtain a projective resolution

$$\cdots \to \bigoplus_{j\in J} P_{j,2} \to \bigoplus_{j\in J} P_{j,1} \to \bigoplus_{j\in J} P_{j,0} \to \bigoplus_{j\in J} B_j \to 0,$$

which, upon tensoring with A and truncating, yields the complex

$$\cdots \to A \otimes_R \bigoplus_{j\in J} P_{j,2} \to A \otimes_R \bigoplus_{j\in J} P_{j,1} \to A \otimes_R \bigoplus_{j\in J} P_{j,0} \to A \otimes_R \bigoplus_{j\in J} B_j \to 0.$$

The homology of this complex is exactly equal to $\operatorname{Tor}_n^R(A, \bigoplus_{j\in J} B_j)$. On the other hand, noting that the tensor product commutes with direct sums, this complex is the direct sum of the complexes

$$\cdots \to A \otimes_R P_{j,2} \to A \otimes_R P_{j,1} \to A \otimes_R P_{j,0} \to A \otimes_R B_j \to 0,$$

whose nth homology module is $\operatorname{Tor}_n^R(A, B_j)$. The fact that homology commutes with direct sums then implies that the nth homology of

$$\cdots \to \bigoplus_{j\in J} A \otimes_R P_{j,2} \to \bigoplus_{j\in J} A \otimes_R P_{j,1} \to \bigoplus_{j\in J} A \otimes_R P_{j,0} \to \bigoplus_{j\in J} A \otimes_R B_j \to 0$$

is exactly $\bigoplus_{j\in J} \operatorname{Tor}_n^R(A, B_j)$. □

Problem 2 (§17.1#15)**.** (a) If I is an ideal in R and M is an R-module, prove that $\mathrm{Tor}_1^R(M, R/I)$ is isomorphic to the kernel of the map $M \otimes_R I \to M$ that maps $m \otimes i$ to mi for $i \in I$ and $m \in M$.

(b) Prove that the R-module M is flat if and only if $\mathrm{Tor}_1^R(M, R/I) = 0$ for every finitely generated ideal I of R.

Solution. (a) The short exact sequence $0 \to I \to R \to R/I \to 0$ is transformed by $\mathrm{Tor}_\bullet^R(M, -)$ into the long exact sequence

$$\cdots \to \mathrm{Tor}_1^R(M, R) \to \mathrm{Tor}_1^R(M, R/I) \to M \otimes_R I \to M \otimes_R R \to M \otimes_R R/I \to 0.$$

There is an isomorphism $M \otimes_R R \cong M$ by sending $m \otimes r \mapsto mr$ (since every $m \otimes r$ can be expresed uniquely as $n \otimes 1$ for some $n \in M$), so the morphism $M \otimes_R I \to M \otimes_R R \to M$ given by composing this isomorphism with the morphism in the exact sequence induced by the inclusion $I \hookrightarrow M$ is the composed map $m \otimes i \mapsto m \otimes i \mapsto mi$.

Since R is a free R-module, it is flat, so $\mathrm{Tor}_1^R(M, R) = 0$. By exactness of the sequence above at $\mathrm{Tor}_1^R(M, R/I)$, this means that $\mathrm{Tor}_1^R(M, R/I)$ maps injectively into $M \otimes_R I$. Then, by exactness at $M \otimes_R I$, the kernel of the $M \otimes_R I \to M \otimes_R R \cong M$ map is equal to the image of the preceding map in the sequence, which we just established is injective, so this image (hence the kernel of the map in question) is isomorphic to $\mathrm{Tor}_1^R(M, R/I)$, as desired.

(b) If M is flat then $\mathrm{Tor}_1^R(M, N) = 0$ for all N.

Conversely, suppose we are given that $\mathrm{Tor}_1^R(M, R/I) = 0$ for every finitely generated ideal $I \subseteq R$; in the long exact sequence for $\mathrm{Tor}_\bullet^R(M, -)$, this means

$$0 \to M \otimes_R I \to M \otimes_R R \to M \otimes_R R/I \to 0$$

is an exact sequence. Flatness then follows from §10.5#25. $\square$

Problem 3 (§17.1#16)**.** Suppose R is a PID and A and B are R-modules. If $t(B)$ denotes the torsion submodule of B, show that $\mathrm{Tor}_1^R(A, t(B)) \cong \mathrm{Tor}_1^R(A, B)$ and deduce that $\mathrm{Tor}_1^R(A, B) \cong \mathrm{Tor}_1^R(t(A), t(B))$.

Solution. Let $B' = t(B)$ and $B'' = B/t(B)$. For any ideal $I = (a)$ of R, the map

$$B'' \otimes_R I \to B'' \otimes_R R \cong B'', \quad m \otimes ra \mapsto mra$$

is injective, since if $m_1 r_1 a = m_2 r_2 a$ then $m_1 r_1 - m_2 r_2$ is annihilated by a, meaning it is zero because $B/t(B)$ is torsionfree, so $m_1 \otimes r_1 a = m_1 r_1 \otimes a = m_2 r_2 \otimes a = m_2 \otimes r_2 a$. Since the kernel of this map is zero, the long exact sequence for $\mathrm{Tor}_\bullet^R(B'', -)$ applied to the short exact sequence $0 \to I \to R \to R/I \to 0$ yields $\mathrm{Tor}_1^R(B'', R/I) = 0$, hence B'' is flat by part (b) of the previous problem. This in turn means that $\mathrm{Tor}_n^R(N, B'') = 0$ for all R-modules N, so the long exact sequence for $\mathrm{Tor}_\bullet^R(A, -)$ applied to $0 \to B' \to B \to B'' \to 0$ looks like

$$\cdots \to \underbrace{\mathrm{Tor}_{n+1}^R(A, B'')}_{=0} \to \mathrm{Tor}_n^R(A, B') \overset{\cong}{\to} \mathrm{Tor}_n^R(A, B) \to \underbrace{\mathrm{Tor}_n^R(A, B'')}_{=0} \to \cdots$$

meaning $\mathrm{Tor}_n^R(A, B) \cong \mathrm{Tor}_n^R(A, t(B)) \cong \mathrm{Tor}_n^R(t(A), t(B))$ for all $n \geq 1$. $\square$

Problem 4 (§17.1#23)**.** Let $D^{-1}R$ be the localization of the commutative ring R with respect to the multiplicative subset D of R. Prove that localization commutes with Tor, i.e.

$$D^{-1}\operatorname{Tor}_n^R(A,B) \cong \operatorname{Tor}_n^{D^{-1}R}(D^{-1}A, D^{-1}B)$$

for all R-modules A and B and all $n \geq 0$.

Solution. By exercise §17.1#22 with $S = D^{-1}R$ and f the obvious inclusion of R into S, we have that $\operatorname{Tor}_n^{D^{-1}R}(D^{-1}A, D^{-1}B) = \operatorname{Tor}_n^R(A, D^{-1}B)$. For a given projective resolution $\cdots \to P_1 \to P_0 \to A \to 0$ of A, $\operatorname{Tor}_\bullet^R(A, D^{-1}B)$ is the homology of

$$\cdots \to P_1 \otimes_R D^{-1}B \to P_0 \otimes_R D^{-1}B \to 0.$$

On the other hand, $-\otimes_R D^{-1}R$ is an exact functor because $D^{-1}R$ is flat, so it preserves kernels and cokernels, hence commutes with homology. Since $D^{-1}B \cong B \otimes D^{-1}R$, we can rewrite the chain complex above as

$$\cdots \to (P_1 \otimes_R B) \otimes_R D^{-1}R \to (P_0 \otimes_R B) \otimes_R D^{-1}R \to 0,$$

and its homology is $\operatorname{Tor}_\bullet^R(A,B) \otimes_R D^{-1}R = D^{-1}\operatorname{Tor}_\bullet^R(A,B)$. □

Problem 5 (§17.1#26)**.**

(a) Prove that every finitely generated module over a Noetherian ring R is finitely presented.

(b) Prove that an R-module M is finitely presented and projective if and only if M is a direct summand of R^n for some integer $n \geq 1$.

Solution. (a) Let M be a finitely generated R-module, say with t generators. Then there is a surjective map $q\colon R^t \to M$ sending each basis element to a generator of M. The kernel of this map must be finitely generated because submodules of finitely generated modules over Noetherian rings are finitely generated. Supposing that $\ker q$ has s generators, there is an exact sequence $R^s \to R^t \xrightarrow{q} M \to 0$ where the leftmost map sends each basis element of R^s to one of the generators of $\ker q$; the existence of this exact sequence is what it means to be finitely presented.

(b) If M is a direct summand of R^n then it is of course projective. The projection $R^n \to M$ shows that M is finitely generated, since it is generated by the images of the basis elements of R^n; (a) then shows that M is finitely presented.

Conversely, suppose M is finitely presented and projective. Then there is a positive integer t and a surjection $R^t \xrightarrow{q} M$ (this is part of the exact sequence we obtain by M being finitely presented). Since M is projective, the identity map $M \to M$ lifts to a section $\sigma\colon M \to R^t$ such that $\sigma \circ q = \mathrm{id}$; therefore, the short exact sequence

$$\begin{tikzcd} 0 \arrow[r] & \ker q \arrow[r] & R^t \arrow[r, "q"] & M \arrow[r] \arrow[l, bend left, dashed, "\sigma"] & 0 \end{tikzcd}$$

splits, by the splitting lemma. □